

OM CHIMURKAR

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 GitHub |  LinkedIn |  Google Scholar |  ResearchGate |  ORCID

Education

Newton School of Technology, Rishihood University
B.Tech in Artificial Intelligence

Expected May 2028
GPA: 8.0 / 10.0

Research & Publications

DKV: Anchor + Low-Rank Differential KV-Cache Compression

Independent Research

IEEE TETCI (Under Review) | *Technical Report & Open-Source Runtime* | *Zenodo: 10.5281/zenodo.21539110*

- Architected a training-free KV-cache compression runtime combining exact anchors, rank-32 joint K|V truncated SVD deltas, and multi-signal exact residual tokens to preserve verbatim recall.
- Achieved $1.44\times$ – $2.25\times$ memory reduction and $1.72\times$ faster prefill scaling at 64k context on Apple M3 (8.6 GB) over memory-optimized dense baselines.
- Engineered MLX and CUDA/Triton backends with autotuned fused decode launches, async H2D prefetching, and in-place residual scattering; exact residual keys prevented needle-recall collapse across 4k–64k contexts with $\Delta\text{PPL} = 0.00\%$.

Deterministic AGI Benchmarking Under Constructed Cognitive Load

Independent Research

Published Preprint | *Zenodo: 10.5281/zenodo.19652443*

- Designed a deterministic benchmark using constructed-language rule-learning tasks to evaluate reasoning without judge-model scoring, reducing judge inflation and memorization effects.
- Developed 116 original tasks across Legal, Science, Social Norms, and Math/Logic; evaluation across seven frontier models revealed a 23.2 percentage-point performance spread.

An Ablation Study of Retrieval Strategies for Stock-News RAG Systems

Independent Research

Published Preprint | *Zenodo: 10.5281/zenodo.19086005*

- Conducted controlled empirical ablations comparing semantic, hybrid (BM25 + vector), and Maximal Marginal Relevance (MMR) retrieval on stock-news QA benchmarks.
- Demonstrated that hybrid retrieval improves coverage over semantic-only search, while MMR increases contextual diversity without proportional grounding gains.

Technical Skills

ML & LLM Systems PyTorch, Hugging Face Transformers, LLM Inference Optimization, KV-Cache Compression, RAG, Model Evaluation, Quantization, Low-Rank Factorization (SVD)

Systems & Tools Python, CUDA, Triton, FastAPI, Git, Vector Databases, Linux

Awards & Experience

Cognizance '25 (SnapSyntax) — IIT Roorkee

1st Place / Zonals Winner

May 2025

- Secured first place in the SnapSyntax technical challenge at IIT Roorkee's annual technical festival.

Google — Gemini API Developer Contest

Contestant

May–August 2024

- Built and deployed an asynchronous LLM application integrating Gemini API with custom prompt pipelines under live competition constraints.

Honors & Certifications

Anthropic: AI Fluency Framework & Foundations

AWS: Fundamentals of Machine Learning & Artificial Intelligence